Estimating the Effect of Activity and Diet on BMI with Causal AI, an Analysis of NHANES Data

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Abstract. Standard machine learning is good at prediction but it cannot say what happens to body mass index when a person changes how they move or what they eat. I treat that as a causal question and build a full pipeline on cross sectional data from the National Health and Nutrition Examination Survey. First I learn candidate graphs with three discovery algorithms that work in different ways, PC, GES, and DirectLiNGAM, and I reconcile their disagreements using bootstrap stability together with a short list of physiological rules. From the resulting graph I read off adjustment sets and estimate the average effect of four exposures on BMI, namely vigorous physical activity and the daily intake of total calories, protein, and carbohydrate. Vigorous activity is linked to a BMI about 1.7 points lower after adjustment, and that effect survives a placebo test, an added synthetic confounder, subsetting, a sensitivity scan, and a double machine learning cross check. The dietary effects are small and depend heavily on how the question is framed, which I show with an energy held substitution analysis. Because the data is a single snapshot, I read every number as a carefully adjusted association rather than proven cause, and I stay explicit about reverse causation and unmeasured confounding throughout.

1 Introduction

Prediction and explanation are not the same job. A model can guess a person's body mass index from a health survey with decent accuracy and still be useless for the question a clinician actually cares about, which is what would happen to that person's BMI if they started exercising or changed their diet. That second question is causal, and a model trained only on correlations answers a different one [1]. Knowing that two things move together does not tell you what moving one of them would do.

In this work I take the causal question seriously and build a pipeline on NHANES, a large cross sectional health survey [2]. The data gives me demographics, body measures, activity levels, and a detailed diet recall for several thousand adults, all recorded at one point in time. I estimate the average effect of four exposures on BMI, doing vigorous physical activity and the daily intake of total calories, protein, and carbohydrate.

The pipeline has two halves. In the first I do not assume I already know the causal structure. I learn it from the data with three discovery algorithms, look at where they agree and where they fall apart, and settle the hard cases with resampling and a short list of physiological facts that no algorithm should be allowed to break. In the second half I use that graph to decide what to control for, estimate the four effects with confidence intervals, and then spend most of the effort trying to knock those estimates down with placebo tests, synthetic confounders, a sensitivity scan, and a double machine learning cross check.

I want to be honest about the ceiling on what any of this can show. The data is one snapshot per person, so for every exposure the arrow could point the other way, and the numbers alone cannot rule that out. I treat the results as adjusted associations that carry a causal reading, not as settled fact, and I say where that reading is weakest.

2 Data and Preparation

The raw file holds 3842 adult records with no obviously empty cells, which turned out to be a little misleading. Two encoding tricks needed fixing before any causal claim would hold up.

The first was a very small float, on the order of 5×10^{-79}, standing in for zero across several columns. A near zero diastolic blood pressure or income ratio is not a real reading, so I set those to missing. The second was the value 9999 in the sedentary minutes column, which is the survey code for a refused or unknown answer rather than someone who sat still for a whole week. I also capped daily energy intake to a plausible adult range of roughly 500 to 5000 kcal, since the 24 hour dietary recall throws off wild outliers that are almost certainly recording errors rather than real diets. Finally I recoded sex into a single female indicator so the models could use it cleanly.

After dropping rows with a missing value in any column I use, 3701 of the 3842 records remain, a loss of 141. This is a complete case analysis, and I come back to what that assumes near the end. Table 1 gives the shape of the working sample. BMI has a mean close to 29.6 and a long right tail, about a quarter of the sample reports vigorous activity, and the macronutrients track total energy closely. That last point is arithmetic rather than a finding, since protein, carbohydrate, and fat add up to calories.

Two variables that look like handy controls are left out of every adjustment set on purpose, and the reason is causal logic rather than model fit. Waist circumference is close to a second way of measuring BMI, so conditioning on it would mean partly conditioning on the outcome itself, which pulls the effects toward zero. Blood pressure sits downstream of body weight, since higher BMI tends to raise it, so it is a descendant of the outcome, and adjusting for a descendant can open non causal paths and spoil the estimate. I also drop total sugars and dietary fiber, because they are parts

Table 1. Working sample after cleaning, 3701 adults.

Variable	Mean	SD	Median	Max
Age (years)	49.35	18.40	50.00	80.00
Female	0.51	0.50	1.00	1.00
Income ratio	2.59	1.60	2.19	5.00
Vigorous activity	0.26	0.44	0.00	1.00
Sedentary (min)	335.1	196.9	300.0	1320
Total calories	2082.6	873.8	1949	4996
Protein (g)	79.02	39.54	71.72	355.7
Carbohydrate (g)	244.7	113.7	226.3	818.8
BMI	29.55	7.13	28.40	67.70

of the carbohydrate total and add redundancy without adding anything causal. Figure 1 shows the outcome and the correlation structure, and both features described above are visible there.

3 Causal Discovery

I ran three algorithms that recover structure in genuinely different ways, and that difference is the point. Agreement between methods that share the same assumptions is cheap. Agreement between methods that do not share them is worth something.

PC is constraint based. It starts fully connected, drops an edge whenever it finds a conditional independence, and orients what the pattern of those independences allows [4]. GES is score based and greedily edits the graph to improve a penalized likelihood, here the BIC. DirectLiNGAM makes a stronger assumption, linear effects with non Gaussian noise, and uses the resulting asymmetry to recover a full ordering of the variables, so unlike the other two it returns a fully directed graph with nothing left undirected.

I ran all three with no background knowledge first, on purpose, so I could see the raw disagreements before feeding in anything I already know. PC returned 14 directed and 3 undirected edges, GES returned 13 directed and 1 undirected, and DirectLiNGAM returned 13 directed edges. Figure 2 places the three graphs side by side.

The methods line up on the easy structure. Demographics come before the lifestyle variables, the three diet variables cluster together, and activity connects to BMI. The disagreements are the useful part. PC points several arrows into age and sex, which is impossible, since nothing a person eats or does can cause their age. This is a known failure of constraint based search when there is no time ordering to anchor direction. GES sidesteps that trap but then argues with LiNGAM about whether calories drive the macronutrients or the macronutrients add up to calories. Both PC and GES also draw an arrow running from BMI into sedentary time, which is reverse causation showing up in graph form, since it is at least as plausible that heavier people sit more as the other way around, and one snapshot cannot separate the two. None of the three graphs is usable as it stands, which is normal for structure learning on data like this, and it is exactly why I compare them rather than crown a winner.

To see which edges are worth trusting, I resampled the rows with replacement and reran PC twelve times, recording how often each connection appeared. Figure 3 shows the result. The edges that turn up in nearly every run are the physiologically sensible ones, the demographic block, the calorie and macronutrient cluster, and the activity to BMI link. The orientations that made no sense are the ones that flicker in and out across runs. So the resampling tells me which parts of the structure the data actually supports and which parts I have to decide myself using prior knowledge.

I then combined what survived the bootstrap with a short list of fixed edges taken from physiology, the background knowledge the task allows me to force in. Nothing may cause age, sex, or income ratio, so every arrow into those three is blocked, with income treated as an upstream marker of socioeconomic position. BMI is the outcome and may not cause any exposure. Protein and carbohydrate point into total calories, because macronutrients add up to energy, and calories, protein, and carbohydrate each keep a separate arrow into BMI so their direct effects stay distinct from the shared path through energy. That last choice does a great deal of work later, because it makes total calories a mediator for the protein and carbohydrate pathways but a confounder when calories is itself the treatment. The final graph, in Figure 4, has 9 nodes and 25 edges and is acyclic by construction.

4 Identification and Estimation

With a graph in hand, identification is close to mechanical. For each exposure I ask the graph which variables block the back door paths into BMI, adjust for exactly that set, and estimate with ordinary least squares. Reading the adjustment sets off the DAG rather than choosing them by hand is the whole reason for drawing it carefully in the first place.

What each number means depends on the exposure, and it is easy to misread them if that is not kept straight. For vigorous activity the back door set is demographics, and the estimate is the average BMI gap between active and inactive people after adjustment. For total calories the graph tells me to also hold the macronutrient composition fixed, because protein and carbohydrate are common causes of both energy and BMI, so the number is the effect of adding energy while composition stays the same. For protein and carbohydrate the back door set is just demographics, and total calories is left out because it is a mediator, so those are total effects that include whatever runs through the extra energy the macronutrient brings with it. I report an energy held version further on, since it answers a different question.

Table 2 collects the four effects. Vigorous activity is the cleanest to read. Being active goes with a BMI about 1.69 points lower, with a 95 percent interval running from roughly -2.25 to -1.13 that sits well clear of zero. Adding 100 kcal per day at fixed composition goes with a BMI about 0.13 higher. The protein effect

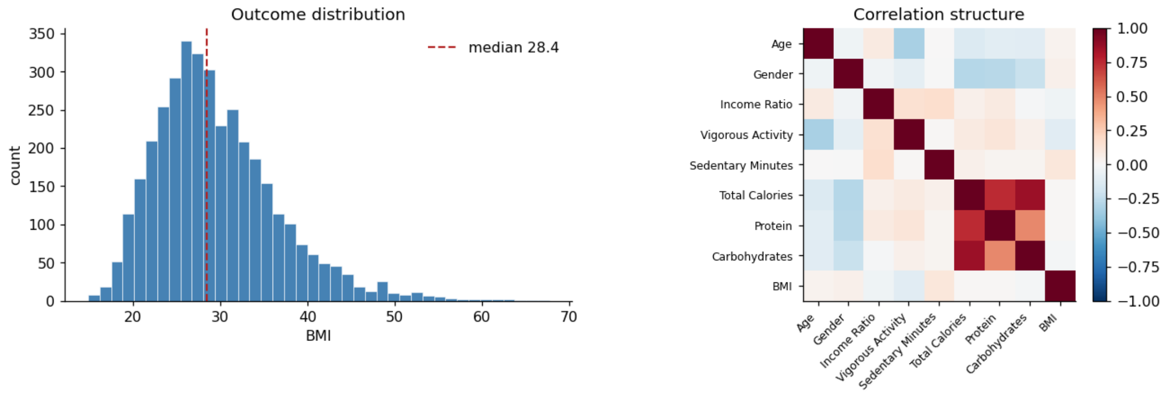

Figure 1. Left, the BMI distribution, which is right skewed with a long high end tail. Right, the correlation structure of the working variables. The strong block linking calories to the macronutrients is arithmetic, not a hidden pattern.

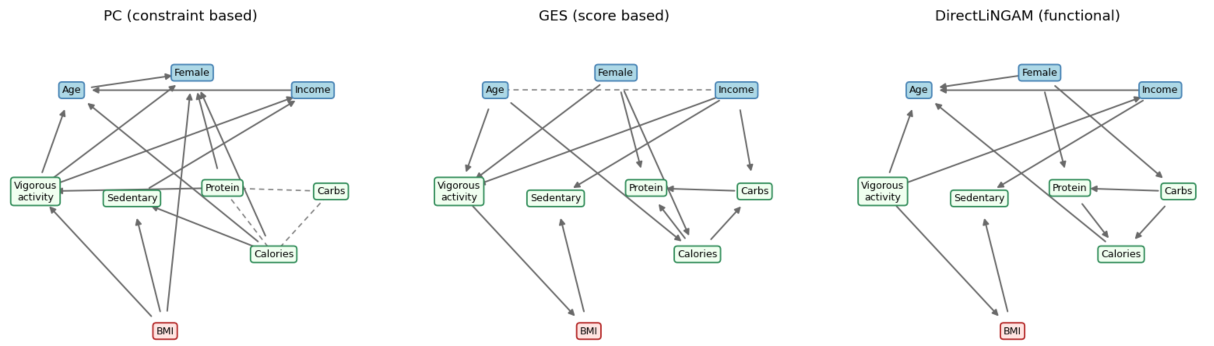

Figure 2. The three discovered graphs with no background knowledge. Dashed lines are undirected edges. Note the arrows pointing into age or sex in the constraint based and score based graphs, which physiology rules out.

Table 2. Estimated average treatment effects on BMI, with 95 percent intervals and the adjustment set read from the graph.

Treatment	Contrast	Effect	95% interval
Vigorous activity	active vs not	−1.689	(−2.25, −1.13)
Total calories	per +100 kcal	0.134	(0.06, 0.21)
Protein	per +10 g	0.060	(−0.00, 0.12)
Carbohydrate	per +10 g	−0.001	(−0.02, 0.02)

Activity, protein, and carbohydrate adjust for age, sex, and income. Calories also holds protein and carbohydrate fixed.

is small and its interval just touches zero, and the carbohydrate effect is essentially nil. Figure 5 shows the same four effects in units that are easier to picture, since per gram and per calorie numbers are tiny by their nature.

Before trusting the activity number I checked it three ways that should agree if the adjustment set is doing its job. Plain regression gives −1.69, DoWhy's own back door regression gives −1.63, and inverse probability weighting, which reweights the sample instead of modeling the outcome, gives −1.55. They land close together, which is what I want to see from estimators that ought to be measuring the same thing.

5 Robustness

An estimate from observational data can never be proven causal, so instead I tried to break it in ways a spurious effect would not survive. I ran the full battery on vigorous activity, the main effect, and let the confidence intervals, the sensitivity scan, and the double ML check carry the robustness load for the dietary exposures, where the same logic applies in the same way.

The first three checks are refutations. A placebo test swaps the real treatment for a shuffled one, and the effect should collapse toward zero. A random common cause adds an invented confounder, and a well identified effect should barely move. A subset check refits on random 85 percent samples and should return a similar number each time if no small group of points is driving the result. The activity effect behaves exactly as a real effect should. The placebo estimate falls to about −0.04, near zero, while the added confounder version stays at −1.63 and the subset version at −1.62, both essentially unchanged from the −1.63 baseline.

These refutations still say nothing about confounders I never measured, which is the real worry with back door adjustment. Genetics, overall diet quality, and non vigorous movement are all absent from this data. So I simulated an unobserved common cause and turned its strength up to see how hard it would have to push to erase the effect. Figure 6 shows the estimate

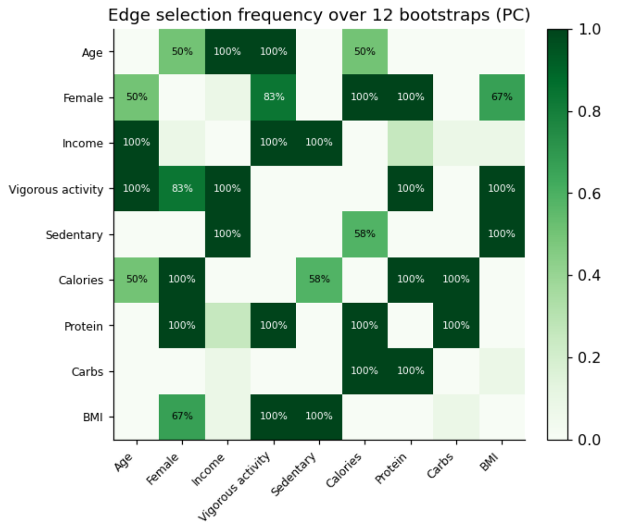

Figure 3. How often each edge appeared across 12 bootstrap reruns of PC. Stable connections match physiology, while the impossible orientations are the unstable ones.

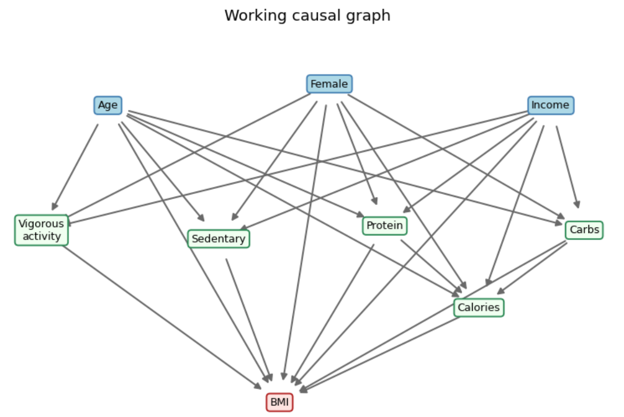

Figure 4. The working causal graph, built from the stable edges plus the fixed physiological rules. Total calories acts as a mediator for the macronutrient pathways and as a confounder when it is the treatment.

staying clearly negative across the range, ending near -1.5 to -1.15 at the strongest setting I tried. That does not rule out confounding, but a hidden cause would have to be implausibly strong to flip the sign of the result.

As a final cross check I re estimated the activity effect with linear double machine learning, which fits flexible gradient boosted models for both the outcome and the treatment and partials them out before estimating, so it does not lean on the linear form the regression assumes. It returns -1.42 with a 95 percent interval from about -1.93 to -0.90, close to the regression value. That agreement tells me the linear model was not quietly doing all the work. Letting the same model vary the effect with age, in Figure 7, gives a fairly flat curve, so I see no strong sign that the activity effect depends on age within this sample, although the bands are wide enough that I would not push the point.

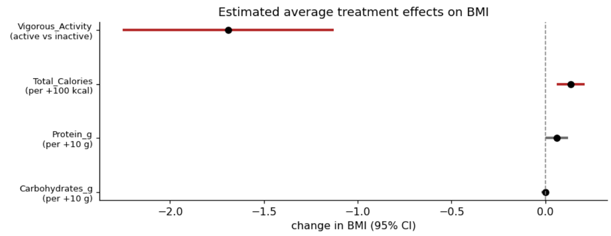

Figure 5. The four estimated effects with 95 percent intervals. Intervals clear of zero are drawn in red. Only vigorous activity and total calories are clearly separated from zero.

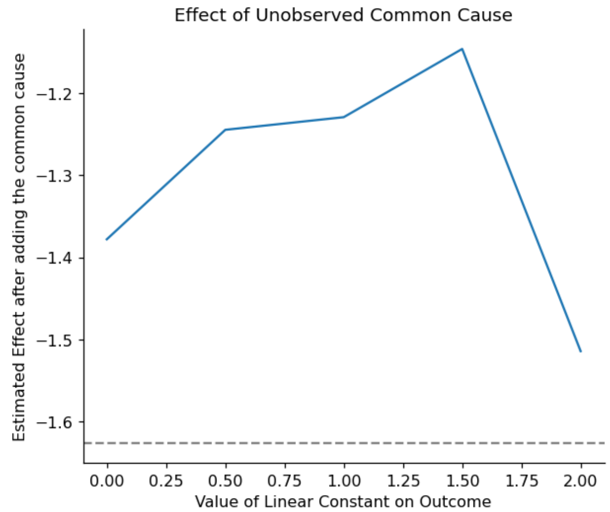

Figure 6. Sensitivity to an unobserved common cause. As the simulated confounder grows stronger the activity effect stays clearly below zero.

The last extension is the one I would actually put in front of a nutritionist. As a plain total effect, eating more protein goes with a slightly higher BMI, about $+0.06$ per 10 g, but that is mostly because people who eat more protein tend to eat more of everything. Once total energy and the other macronutrient are held fixed, so the comparison becomes a genuine swap of protein for other energy, the sign flips to about -0.05, and the same energy held comparison for carbohydrate gives about -0.08. Both numbers are small and neither one settles anything, but they answer two different questions, and treating a total effect as if it were a substitution effect is a common way nutrition claims go wrong.

6 Limitations

The design is the main limitation. The data is cross sectional, so every exposure could plausibly run backward. Heavier people may sit more, eat differently, and move less because of their weight rather than the other way around, and no amount of adjusting repairs a reversed arrow. Longitudinal data, with exposures measured before the outcome, would change what can honestly be claimed here.

The diet figures come from a single 24 hour re-

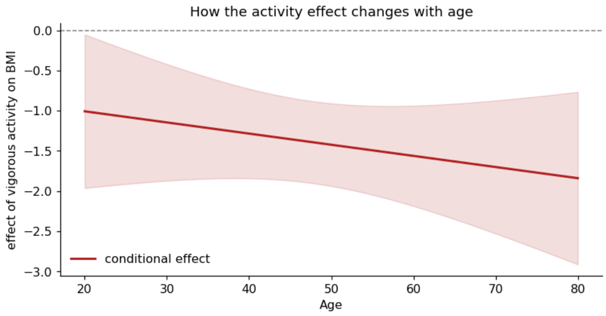

Figure 7. The activity effect on BMI as it varies with age, from double machine learning. The curve is close to flat, with wide uncertainty bands.

call, which is noisy, tends to under report, and is a thin proxy for someone's usual eating. That kind of measurement error usually shrinks effects toward zero, so the small dietary numbers may be understated rather than truly negligible. I also did not apply the NHANES survey weights, so the sample does not represent the wider US population and the estimates describe this specific sample rather than the country. A production version of the analysis would bring in the weights and the survey design variables.

Finally, back door adjustment only works if the major confounders are measured, and several are not, among them genetics, medications, and overall diet pattern. The sensitivity scan suggests the activity effect is not fragile, but there is enough room left for unmeasured confounding that every number here should be read as a carefully adjusted association carrying a causal interpretation, not as a closed causal fact.

7 Conclusion

I built a causal pipeline end to end on NHANES, reconciled three discovery methods into a single graph I can defend, took the adjustment sets from that graph instead of guessing them, and estimated four effects on BMI with intervals and several stress tests. Vigorous activity is the result I would stand behind most firmly, a lower BMI of roughly 1.5 to 1.7 points that held up under every check I ran. The dietary effects are smaller, more sensitive to how the question is framed, and more exposed to the weaknesses of a single cross sectional snapshot, and I think stating that plainly is part of doing the analysis honestly.

References

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[4] A. Zanga, E. Ozkirimli, and F. Stella. A survey on causal discovery, theory and practice. *International Journal of Approximate Reasoning*, vol. 151, pp. 101–129, 2022.